Independent Mathematical Review Requested

The Mirror Cube, the Self-Referential Unit, and P versus NP

Le Matt Ansatz Di Ego · Relational Geometry Program · October 30, 2026
Repository: https://github.com/EndlessMonkeyProyect/programa-geometria-relacional

We invite independent assessment of a research program that begins with a relational notion of a self-referential unit and tests its mathematical realizations in reflection groups, modular normalization, graph interfaces, and 3-coloring. A **relational radius** specifies the reach of the relations sustaining a unit; contextual uncertainty refers to distinctions not determined under a given reference and query. This is the program's ontological motivation, not a premise of the mathematical proofs.

What can be reviewed independently:

- **O59 / CME:** exact orbits of two reflections, a counterexample to updating a folded signature after moving a mirror, and classical Coxeter-group growth classifications.
- **MF-97 / MF-106:** observational equivalence under continuations and the exact criterion for updating a quotient signature; mathematical existence is distinguished from computational efficiency.
- **R-01, R4, R5, R6:** modular reference refinement, locally closed tensions with global periods, composition through graph boundaries, and contextual novelty of graph constraints.
- **R-04:** diamond closure, determination radius, and the coexistence of growing support with short refutation certificates on specific Mycielski families.
- **T-EXT-01:** for labeled bipartite graphs on n vertices, a full-interface signature answering every five-edge continuation D_{uv} must distinguish $2^{\lfloor n^2/4 \rfloor}$ states, requiring at least $\lfloor n^2/4 \rfloor$ bits in a fixed-length encoding. The bound is tight on this family, quadratic rather than superpolynomial, and is **not** a lower bound on general 3-coloring time. Reported checks cover 4,672 exact queries.

What we are asking: Please examine the hypotheses, proofs, reproducibility, bibliographic precedents, and possible stronger formulations. In particular, can an internal reference be made nontrivially distinct from an exterior signature? Which proof systems simulate explicitly defined relational procedures, and which lower bounds transfer? Does T-EXT-01 follow from a known contextual-equivalence or communication-complexity theorem?

Status: The program does **not** claim $P = NP$ or $P \neq NP$. The cited N1-B simulation has not been independently revalidated in this edition. The full request contains 13 independent reviewer questions, a detailed proof of T-EXT-01, and provenance for historical verifiers.

Please reply through an issue in the repository or the publication thread, quoting the question or statement under review. A rigorous partial response, counterexample, corrected proof, or reference is welcome.